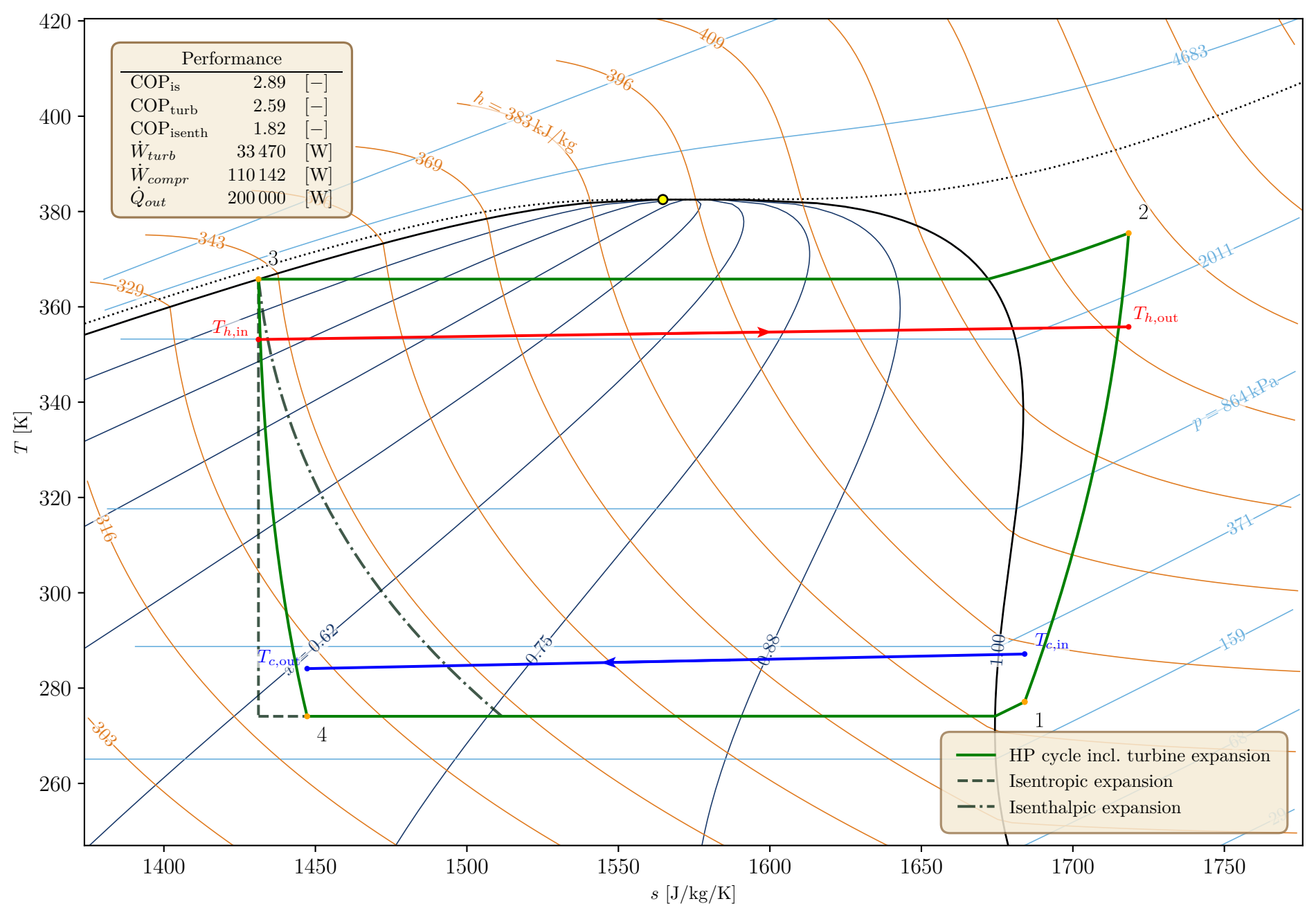

Performance
COP_is 2.89 [–]
COP_turb 2.59 [–]
COP_isenth 1.82 [–]
Ẇ_turb 33 470 [W]
Ẇ_compr 110 142 [W]
Q̇_out 200 000 [W]
h = 383 kJ/kg
409
396
369
343
329
316
303
4683
2011
p = 864 kPa
371
159
68
20
0.62
0.75
0.88
100
T_h,in
T_h,out
T_c,in
T_c,out
1
2
3
4
HP cycle incl. turbine expansion
Isentropic expansion
Isenthalpic expansion
T [K]
s [J/kg/K]
420
400
380
360
340
320
300
280
260
1400
1450
1500
1550
1600
1650
1700
1750